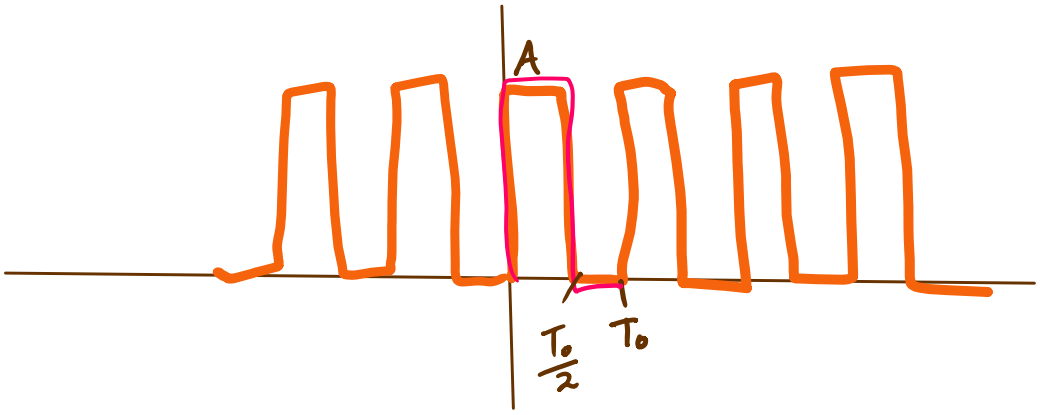
Fourier Series Calculation

Monday, October 5, 2026 11:17 AM



$$\int e^{pt}dt = \frac{e^{pt}}{p}$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t} \qquad \Omega_0 = \frac{2\pi}{T_0} \Rightarrow \Omega_0 T_0 = 2\pi$$

$X_k = ?$

$$X_k = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jk\Omega_0 t}dt = \frac{1}{T_0}\int_0^{T_0/2} A e^{-jk\Omega_0 t}dt$$

$p = -jk\Omega_0$

$$= \frac{A}{T_0}\cdot\frac{1}{-jk\Omega_0} e^{-jk\Omega_0 t}\Big|_0^{T_0/2}$$

$$= \frac{A}{T_0}\cdot\frac{1}{-jk\Omega_0}\left(e^{-jk\Omega_0\cdot\frac{T_0}{2}} - e^{0}\right) = -\frac{A}{jT_0k\Omega_0}\left[e^{-jk\frac{2\pi}{2}} - 1\right]$$

$$X_k = \frac{A}{-jk\cdot 2\pi}\left[e^{-jk\pi} - 1\right] \qquad [\cos(-\theta) = \cos\theta]$$

$$e^{-jk\pi} = (-1)^k = \cos(-k\pi) + j\sin(-k\pi)$$

$$= (-1)^k + j\cdot 0 = (-1)^k$$

If we look at the expression for X_k

for even k, $X_k = 0$ why? because $(-1)^k = 1$ for $k = 2m$, $m \neq 0$

So $X_k = \frac{A}{-jk2\pi}[1-1] = 0$

for odd k, $k = 2m+1$

$$X_k = \frac{A}{-j k 2\pi}\left[-1-1\right] = \frac{-A}{j(2m+1)2\pi}\times -2$$

$$= \frac{A}{j(2m+1)\pi}$$

for $k=0$

$$X_0 = \frac{1}{T_0}\int_0^{T_0} x(t)dt = \frac{1}{T_0}\int_0^{T_0/2} A\,dt$$

$$= \frac{A}{2}$$

$$X_k = \begin{cases} 0 & \text{for } k \text{ even}, k\neq 0 \\ \frac{A}{jk\pi} & \text{for } k \text{ odd}, k\neq 0 \\ \frac{A}{2} & \text{for } k=0 \end{cases}$$

$$x(t) = \frac{A}{2} + \sum_{m=-\infty}^{m=\infty} \frac{A}{j(2m+1)\pi} e^{j(2m+1)\Omega_0 t}$$

→ we used $k=2m+1$ for odd part

$$x(t) = \frac{A}{2} + \frac{A}{j\pi}\sum_{m=-\infty}^{m=+\infty} \frac{e^{j(2m+1)\Omega_0 t}}{2m+1}$$

If we choose $m = 1000$ or any finite number then let's call such approx. $\tilde{x}(t)$

we might want

$$\sqrt{\frac{1}{N}\sum \left| x(t) - \tilde{x}(t) \right|^2} < \varepsilon$$

RMS Root Mean Square

ε ↓ Some threshold

(b) Similarly we can do for Trigonometric Fourier Series

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$$x(t) = X_0 + 2\sum_{k=1}^{\infty}\left[C_k \cos(k\Omega_0 t) + d_k \sin(k\Omega_0 t)\right]$$

$X_0 = C_0$

General formula for Trigonometric Fourier Series ←

$$C_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\cos(k\Omega_0 t)\,dt \qquad k=0,1,\ldots$$

$$d_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\sin(k\Omega_0 t)\,dt \qquad k=1,2,\ldots$$

Coefficient for Complex exponential fourier series ↙

$$X_k = \sqrt{C_k^2 + d_k^2}$$

$$\theta_k = -\tan^{-1}\left[\frac{d_k}{C_k}\right]$$

$t_0 = 0$ for Convenience

$$C_k = \frac{1}{T_0}\int_0^{T_0/2} \ldots + \underbrace{\int_{T_0/2}^{T_0} \ldots}_{=0}$$

$$C_k = \frac{1}{T_0}\int_0^{T_0/2} A\cdot\cos(k\Omega_0 t)\,dt$$

$$= \frac{1}{T_0}\left.\frac{A\sin(k\Omega_0 t)}{k\Omega_0}\right|_0^{T_0/2}$$

$$\int \cos(ax)\,dx = +\frac{\sin(ax)}{a}$$

$$= \left.\frac{A\sin(k\Omega t)}{2\pi k}\right|_0^{T_0/2} = \frac{A}{2\pi k}\left[\sin\left(\frac{T_0}{2}\Omega_0 k\right) - \sin(0)\right]$$

$$= \frac{A}{2\pi k}\left(\sin\left(\frac{2\pi k}{2}\right) - \sin(0)\right)$$

$$= 0$$

$$d_k = \frac{1}{T_0}\int_0^{T_0/2} A\sin(k\Omega_0 t)\,dt$$

$$= -\frac{1}{\ldots} A\cos(k\Omega_0 t)\,\Big|^{T_0/2}$$

$$= -\frac{1}{T_0} A \frac{\cos(k\Omega_0 t)}{k\Omega_0} \Big|_0^{T_0/2}$$

$$= -\frac{1}{2\pi k} A \left[\cos\left(\frac{T_0}{2} \cdot \Omega_0 k\right) - \cos(0) \right]$$

$$= -\frac{A}{2\pi k} \left[\cos(\pi k) - 1 \right]$$

$$d_k = -\frac{A}{2\pi k} \left[(-1)^k - 1 \right]$$

$k = 2m \Rightarrow$ even $\qquad d_{2m} = 0$

$k = 2m+1 \Rightarrow$ odd

$$d_{2m+1} = \frac{-A}{2\pi(2m+1)} \left[(-1)^{2m+1} - 1 \right]$$

$$= \frac{-A \cdot (-2)}{2\pi(2m+1)} = \frac{A}{\pi(2m+1)}$$

$$x(t) = \frac{A}{2} + \frac{2 \cdot A}{\pi} \sum_{m=0}^{\infty} \frac{1}{(2m+1)} \sin\left((2m+1)\Omega_0 t\right)$$

$$= \frac{A}{2} + \frac{2A}{\pi} \left[\sin(\Omega_0 t) + \frac{1}{3}\sin(3\Omega_0 t) + \frac{1}{5}\sin(5\Omega_0 t) + \cdots \right]$$

